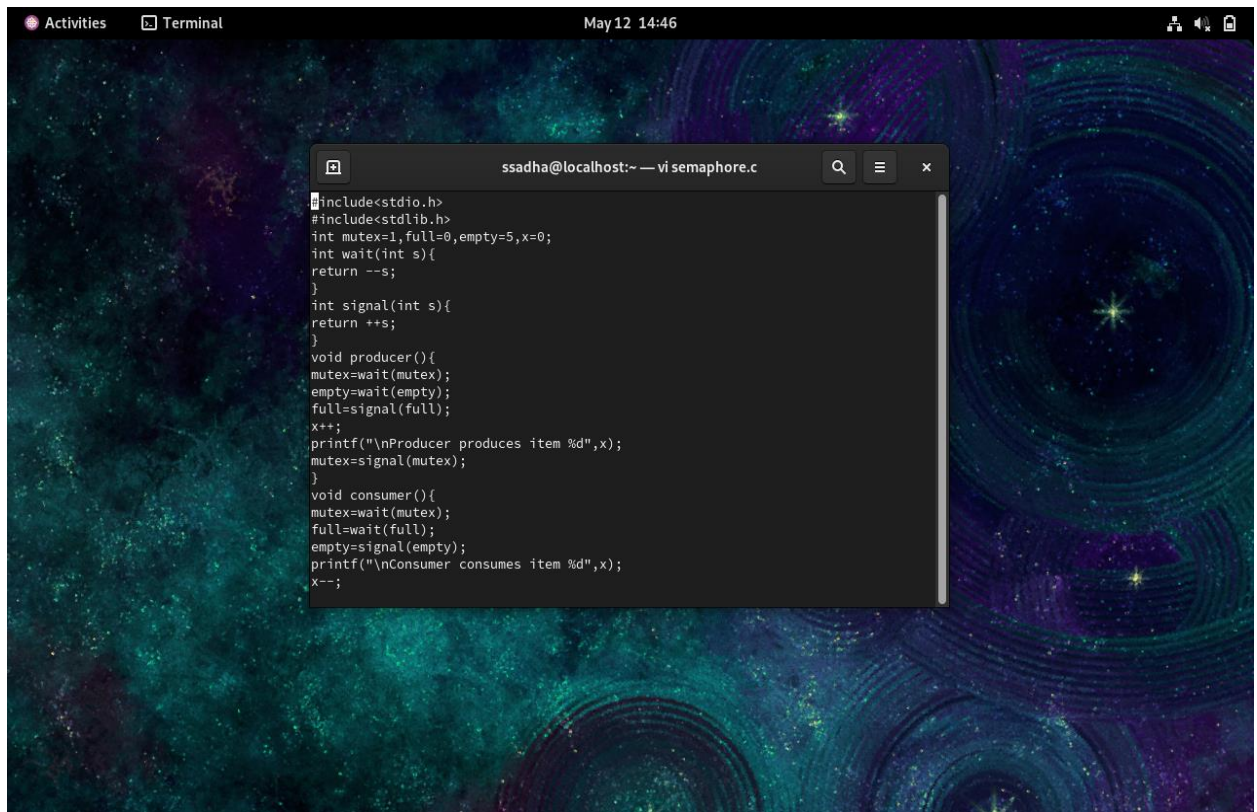
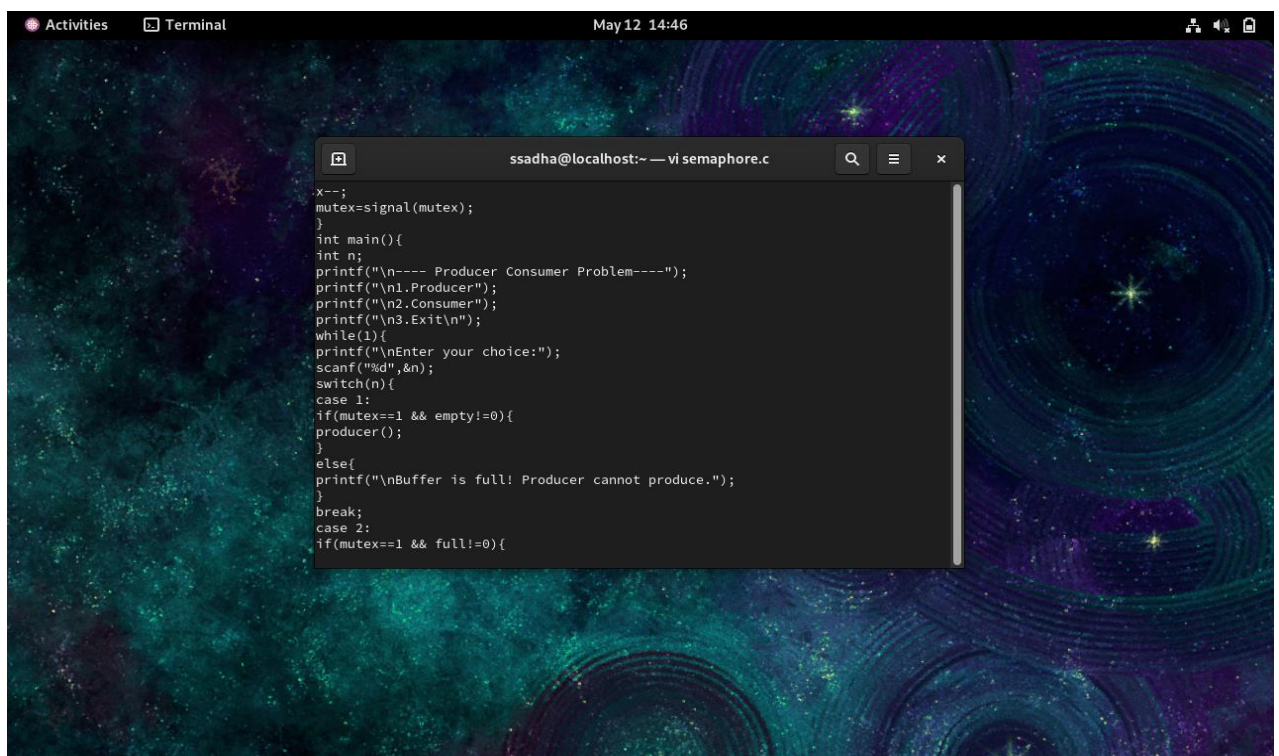


Expt 8: Producer Consumer Using Semaphores:



A screenshot of a terminal window titled "ssadha@localhost:~ — vi semaphore.c". The code displayed is the first part of a C program for the Producer-Consumer problem using semaphores. It includes headers, initializes variables (mutex=1, full=0, empty=5, x=0), and defines wait and signal functions. It also defines the producer and consumer functions, which use these semaphores to manage the production and consumption of items.

```
#include<stdio.h>
#include<stdlib.h>
int mutex=1,full=0,empty=5,x=0;
int wait(int s){
return --s;
}
int signal(int s){
return ++s;
}
void producer(){
mutex=wait(mutex);
empty=wait(empty);
full=signal(full);
x++;
printf("\nProducer produces item %d",x);
mutex=signal(mutex);
}
void consumer(){
mutex=wait(mutex);
full=wait(full);
empty=signal(empty);
printf("\nConsumer consumes item %d",x);
x--;
```



A screenshot of a terminal window titled "ssadha@localhost:~ — vi semaphore.c". The code displayed is the second part of the C program, including the main function. It prompts the user to enter a choice (1 for producer, 2 for consumer) and uses a switch statement to call the producer or consumer function based on the input. It also includes a break statement to exit the loop.

```
x--;
mutex=signal(mutex);
}
int main(){
int n;
printf("\n---- Producer Consumer Problem----");
printf("\n1.Producer");
printf("\n2.Consumer");
printf("\n3.Exit\n");
while(1){
printf("\nEnter your choice:");
scanf("%d",&n);
switch(n){
case 1:
if(mutex==1 && empty!=0){
producer();
}
else{
printf("\nBuffer is full! Producer cannot produce.");
}
break;
case 2:
if(mutex==1 && full!=0){
```

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```
ssadha@localhost:~ — vi semaphore.c
producer();
}
else{
printf("\nBuffer is full! Producer cannot produce.");
}
break;
case 2:
if(mutex==1 && full!=0){
consumer();
}
else{
printf("\nBuffer is empty! Consumer cannot consume.");
}
break;
case 3:
printf("\nExiting program..");
printf("\nFinal buffer items produced. %d\n",x);
exit(0);
default:
printf("\nInvalid choice. Try again.");
}
}
```

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```
ssadha@localhost:~
[ssadha@localhost ~]$ vi semaphore.c
[ssadha@localhost ~]$ gcc semaphore.c
[ssadha@localhost ~]$ gcc semaphore.c -o semaphore
[ssadha@localhost ~]$ ./semaphore

---- Producer Consumer Problem----
1.Producer
2.Consumer
3.Exit

Enter your choice:1

Producer produces item 1
Enter your choice:1

Producer produces item 2
Enter your choice:2

Consumer consumes item 2
Enter your choice:3

Exiting program..
Final buffer items produced. 1
[ssadha@localhost ~]$
```